SAD

Segment 3

Aut-23

3.a) What is the significance of feasibility analysis in project management? Which phase of project management requires feasibility analysis? Briefly discuss about three major areas to conduct a feasibility study of an organization.

Feasibility analysis is the cornerstone of effective project management because it determines whether a proposed project is worth doing. The feasibility study allows us to preview the potential outcome of a project and to decide whether it should be continued.

A thorough feasibility study examines a proposed project from several angles to ensure all aspects of the business including technical and financial landscapes are considered. Three major areas for conducting a feasibility study are:

1. **Technical Feasibility:**
    - The technical feasibility assessment is focused on gaining an understanding of the present technical resources of the organization and their applicability to the expected needs of the proposed system.
    - It is an evaluation of the hardware and software and how it meets the need of the proposed system.

2. **Economic Feasibility:**
    - The purpose of the economic feasibility assessment is to determine the positive economic benefits to the organization that the proposed system will provide.
    - It includes quantification and identification of all the benefits expected. This assessment typically involves a cost/ benefits analysis.
3. **Operational Feasibility:**
   Operational feasibility refers to the measure of solving problems with the help of a new proposed system. It helps in taking advantage of the

opportunities and fulfills the requirements as identified during the development of the project. It takes care that the management and the users support the project.

b) Describe the steps to conduct the economic feasibility analysis with proper example.

Or

Define the following:
i) Deliverables
ii) Break-Even Point (with mathematical explanation)
iii) Net Present Value (with mathematical explanation)
iv) Recurring cost
v) Return on Investment (with mathematical explanation)

Steps to conduct economic feasibility:

1. Identify Costs and Benefits

The first task is to identify the costs and benefits of the system. The costs and benefits can be broken down into four categories:

1. Development costs
    - Development costs are those tangible expenses that are incurred during the creation of the system, such as salaries for the project team.
    - Development costs are usually thought of as one-time costs.

2. Operational costs
    - Operational costs are those tangible costs that are required to operate the system, such as the software licensing fees.
    - Operational costs are usually thought of as ongoing costs.

3. Tangible benefits
    - Tangible benefits include revenue that the system enables the organization to collect, such as increased sales. In addition, if the

system produces a reduction in needed staff, lower salary costs result. So, the reduction in costs also is a tangible benefit of the new system.

4. Intangible benefits
    - Intangible costs and benefits are more difficult to incorporate into the economic feasibility analysis because they are based on intuition and belief rather than on "hard numbers." Ex: Public relation, recognition, improved customer service etc
    - Nonetheless, they should be listed in the spreadsheet along with the tangible items.

Other costs are One-Time Cost (initiation and development costs ex: site preparation, software purchases etc.) and Recurring (Operational) Costs (Associated with on-going use of the system ex: software maintenance, incremental data storage expense, incremental communication etc.)

2. Assign Values to Costs and Benefits

- After identifying costs and benefits, the analyst must assign dollar values. Even though it's difficult to predict future outcomes, it is somewhat possible to give an assumption. Only then can the approval committee make an informed decision about whether or not to move ahead with the project.
- The most effective strategy for estimating costs and benefits is to rely on the people who have the best understanding of them (business users and IT professionals). Even intangibles should be valued if at all possible.
- With a likelihood (probability) estimate to each value, an expected value for the cost or benefit can be calculated.
- Intangible benefits and costs are also measurable without assigning a dollar value. It is suggested that organization can quantify intangible costs or benefits (if at all possible).

3. Determine Cash Flow

A formal cost–benefit analysis usually contains costs and benefits over a selected number of years (usually, 3 – 5years) to show cash flow over time.

4. Assess Project's Economic Value

Evaluate the project's expected returns in comparison to its costs using one or more of the following evaluation techniques:

1. Determine ROI

$$ROI = \frac{Total\ Revenue - Total\ Cost}{Total\ Cost}$$

2. Determine BEP

$$\text{BEP} = \begin{matrix}\text{Number of}\\ \text{years of}\\ \text{negative}\\ \text{cash flow}\end{matrix} + \frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$$

3. Determine NPV

$$NPV = \sum PV\ of\ Total\ Benefits - \sum PV\ of\ Total\ Costs$$

OR

i. **Deliverable:** is something that the project produces, and usually happens at certain intervals in the project (called milestones). For example, a deliverable in a software development project can be "creation of the database".

ii. **Break-Even Point:** The break-even point (also called the payback method) is defined as the number of years it takes a firm to recover its original investment in the project from net cash flows.

$$\text{BEP} = \begin{matrix}\text{Number of}\\ \text{years of}\\ \text{negative}\\ \text{cash flow}\end{matrix} + \frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$$

iii. **Net Present Value:** The NPV is simply the difference between the total present value of the benefits and the total present value of the costs.

$$NPV = \sum PV\ of\ Total\ Benefits - \sum PV\ of\ Total\ Costs$$

iv. **Recurring Cost:** is associated with on-going use of the system

- New human resource costs
- Application software maintenance
- Incremental data storage expense
- Incremental communications
- New software and hardware releases
- Consumable supplies

v. **Return on Investment:** The return on investment (ROI) is a simple calculation that divides the project's net benefits (total benefits - total costs) by the total costs. The ROI formula is:

$$ROI = \frac{Total\ Revenue - Total\ Cost}{Total\ Cost}$$

A high ROI suggests that the project's benefits far outweigh the project's cost. ROI is commonly used in practice; however, it is hard to interpret and should not be used as the only measure of a project's worth.

Spring-24

3. a) Define project management. Describe the execution phase of project management.

A project is a temporary endeavor designed to produce a unique product, service or result with a defined beginning and end (usually time-constrained, and often constrained by funding or deliverables) undertaken to meet unique goals and objectives, typically to bring about beneficial change or added value.

The execution phase is the 3rd phase of project management.

Project execution puts the baseline project plan into action. Within the context of the SDLC, project execution occurs primarily during the analysis, design, and implementation phases. There are five project planning activities:

1. **Executing the baseline project plan:** As project manager, you oversee the execution of the baseline plan – that is,
   - you initiate the execution of project activities,
   - acquire and assign resources,
   - orient and train new team members,
   - keep the project on schedule,
   - and ensure the quality of project deliverables.
2. **Monitoring project progress against the baseline project plan:** While you execute the baseline project plan, you should monitor your progress. If the project gets ahead of (or behind) schedule, you may have to adjust resources, activities, and budgets. Monitoring project activities can result in modifications to the current plan.
3. **Managing changes to the baseline project plan:** You will encounter pressure to make changes to the baseline plan. It is recommended that only approved changes to the project specification can be made, and all changes must be reflected in the baseline plan and project workbook, including all charts.
4. **Maintaining the project workbook:** Maintaining complete records of all project events is necessary. The workbook provides the documentation new team members require to assimilate project tasks quickly. It explains why design decisions were made and is a primary source of information for producing all project reports.

5. **Communicating the project status:** The project manager is responsible for keeping all team members – system developers, managers, and customers – abreast of the project status. Clear communication is required to create a shared understanding of the activities and goals of the project; such an understanding ensures better coordination of activities. This means that the entire project plan should be shared with the entire project team, and any revisions to the plan should be communicated to all interested parties so that everyone understands how the plan is evolving.

b) A very basic cash flow projection for an IT Projects is, in the Year 0,1,2,3 costs are 50000, 8000, 10000, 14000 and benefits are 0, 25000, 30000, 48000. Find out the followings for the above conditions:

(i) Return on Investment

(ii) Break-Even Point

OR (of 3b)

Describe cost-benefit analysis. Briefly describe the different metrics to measure projects feasibility in terms of economic value.

Year	Year 0	Year 1	Year 2	Year 3	Total ~~Year 4~~
① Total Benefits	0	25,000	30,000	48,000	103,000
② Total Costs	50,000	8,000	10,000	14,000	82,000
③ Net Benefits (① - ②)	(50,000)	17,000	20,000	34,000	21,000
④ Cumulative Net Cash flow	(50,000)	~~(25000)~~ (33,000)	~~5,000~~ (13,000)	~~53,000~~ $21,000	

$$ROI = \frac{\text{Total Rev.} - \text{Total Cost}}{\text{Total cost}}$$

$$= \frac{103,000 - 82,000}{82,000}$$

$$= 0.256$$

$$BEP = 2 + \frac{34,000 - 21,000}{34,000}$$

$$= 2.382 \text{ years}$$

OR

Cost-benefit analysis or Economic feasibility is the economic feasibility analysis of a project.

- The purpose of the economic feasibility assessment is to determine the positive economic benefits to the organization that the proposed system will provide.
- It includes quantification and identification of all the benefits expected. This assessment typically involves a cost/ benefits analysis.

IT projects commonly involve an initial investment that produces a stream of benefits over time, along with some ongoing support costs. Therefore, the value of the project must be measured over time. Cash flows, both inflows and outflows, are estimated over some future period. Then, these cash flows are evaluated using several techniques to judge whether the projected benefits justify incurring the costs.

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3.	a)	Discuss both internal and external factors that affect system project.	5	CO1
	b)	Explain the role of SWOT analysis in the strategic planning process for IT projects. How does a SWOT analysis help IT managers in identifying strengths, weaknesses, opportunities, and threats? Provide examples of how each element can impact IT project decisions	5	CO3
		<u>OR</u> (of 3b)		
		Describe the steps to conduct the economic feasibility analysis with proper example.	5	